Machine Intelligence 1 Tutorial

04 Gradient methods for parameter optimization

Fachgebiet Neuronale Informationsverarbeitung (NI)

WiSe 23/24

Linear neuron for regression: analytical solution

- Data, Model and Objective
- Calculating the gradient
- Finding minima
- Represent the cost, gradient and Hessian in matrix form
- Solution with regularization

Gradient methods

- The need for gradient-based optimization
- "Scope" of the gradient
- Learning rate schedules
- The Conjugate gradient method
  - Motivation

References

The setting for our linear regression

Data:

observations	$x \in \mathbb{R}$
label	$y_T \in \mathbb{R}$
Dataset	$\left\{ \left(x^{(1)}, y_T^{(1)}\right), \ldots, \left(x^{(\alpha)}, y_T^{(\alpha)}\right), \ldots, \left(x^{(p)}, y_T^{(p)}\right) \right\}$

Model:
linear neuron:

$$y(x; \underline{\mathbf{w}}) = w_0 + w_1 x = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}$$

with $\underline{\mathbf{x}} := (1, x)^\top \in \mathbb{R}^2$ and $\underline{\mathbf{w}} := (w_0, w_1)^\top \in \mathbb{R}^2$

Cost function:

Quadratic error

$$E^T = \frac{1}{p} \sum_{\alpha=1}^{p} \frac{1}{2} \left(y(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}) - y_T^{(\alpha)} \right)^2$$
$$= \frac{1}{p} \sum_{\alpha=1}^{p} e^{(\alpha)}$$

Optimization:

Cost function:
Quadratic error

$$E^T = \frac{1}{p}\sum_{\alpha=1}^{p} \frac{1}{2}\left(y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}}) - y_T^{(\alpha)}\right)^2$$
$$= \frac{1}{p}\sum_{\alpha=1}^{p} e^{(\alpha)}$$

Optimization:

$$E^T \stackrel{!}{=} \min_{\underline{\mathbf{w}}}$$
$$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} = \underline{\mathbf{0}} \qquad \text{(satisfied by extrema)}$$

Calculating the gradient

$$e = \frac{1}{2}\,(y - y_T)^2$$

$$y = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}$$

$$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} = \frac{1}{p}\sum_{\alpha=1}^{p} \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}$$

$$\frac{\partial e}{\partial \underline{\mathbf{w}}} \overset{\text{chain rule}}{=} \frac{\partial e}{\partial y} \cdot \frac{\partial y}{\partial \underline{\mathbf{w}}}$$

$$= \left(y - y_T\right) \cdot \frac{\partial}{\partial \underline{\mathbf{w}}} \underline{\mathbf{w}}^\top \underline{\mathbf{x}}$$

using $\frac{\partial}{\partial \underline{\mathbf{a}}}\left(\underline{\mathbf{a}}^\top \underline{\mathbf{b}}\right) = \underline{\mathbf{b}} = (b_1, b_2, \ldots, b_N)^\top$

$$= \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} - y_T\right) \cdot \underline{\mathbf{x}} \quad = \quad \underline{\mathbf{x}}\,\underline{\mathbf{w}}^\top \underline{\mathbf{x}} - \underline{\mathbf{x}}\, y_T$$

$$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} = \frac{1}{p}\sum_{\alpha=1}^{p} \left(\underline{\mathbf{x}}^{(\alpha)}\,\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}\, y_T\right) \quad =: \quad \underline{\mathbf{g}}$$

Finding minima

$$\underline{\mathbf{H}} \equiv \frac{\partial^2 E^T}{\partial \underline{\mathbf{w}}^2} = \begin{pmatrix} \frac{\partial^2 E^T}{\partial w_0^2} & \frac{\partial^2 E^T}{\partial w_0 \partial w_1} & \cdots & \frac{\partial^2 E^T}{\partial w_0 \partial w_N} \\ \frac{\partial^2 E^T}{\partial w_1 \partial w_0} & \frac{\partial^2 E^T}{\partial w_1^2} & \cdots & \frac{\partial^2 E^T}{\partial w_1 \partial w_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 E^T}{\partial w_N \partial w_0} & \frac{\partial^2 E^T}{\partial w_N \partial w_1} & \cdots & \frac{\partial^2 E^T}{\partial w_N^2} \end{pmatrix}$$

with $\underline{\mathbf{g}} = \frac{\partial E^T}{\partial \underline{\mathbf{w}}}$

Finding minima

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with $\underline{\mathbf{g}} = \frac{\partial E^T}{\partial \underline{\mathbf{w}}}$ the i-th <u>row</u> in $\underline{\mathbf{H}}$ resembles $\frac{\partial}{\partial w_i} \underline{\mathbf{g}}^\top$.
Therefore,

$$\underline{\mathbf{H}} := \frac{\partial}{\partial \underline{\mathbf{w}}} \underline{\mathbf{g}}^\top$$

$$\underline{\mathbf{H}} := \frac{\partial}{\partial \underline{\mathbf{w}}} \underline{\mathbf{g}}^\top = \frac{1}{p} \sum_{\alpha=1}^{p} \frac{\partial}{\partial \underline{\mathbf{w}}} \left(\frac{e^{(\alpha)}}{y(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})} \cdot \frac{\partial y(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})}{\partial \underline{\mathbf{w}}} \right)^\top$$

$$= \frac{1}{p} \sum_{\alpha=1}^{p} \frac{\partial}{\partial \underline{\mathbf{w}}} \left(\underline{\mathbf{x}}^{(\alpha)} \, \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)} \right)^\top$$

using $\left(\underline{\mathbf{A}} \; \underline{\mathbf{B}} \; \underline{\mathbf{C}}\right)^\top = \underline{\mathbf{C}}^\top \underline{\mathbf{B}}^\top \underline{\mathbf{A}}^\top$ and $\underline{\mathbf{x}}^{(\alpha)\top} \underline{\mathbf{w}} = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)}$

$$= \frac{1}{p} \sum_{\alpha=1}^{p} \left(\frac{\partial}{\partial \underline{\mathbf{w}}} \, \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} \; \underline{\mathbf{x}}^{(\alpha)\top} - \frac{\partial}{\partial \underline{\mathbf{w}}} \Big(\underbrace{\underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)}}_{\text{indep. of } \underline{\mathbf{w}}} \Big)^\top \right)$$

using the product rule

$$\underline{\mathbf{H}} = \frac{1}{p} \sum_{\alpha=1}^{p} \underline{\mathbf{x}}^{(\alpha)} \; \underline{\mathbf{x}}^{(\alpha)\top}$$

For the quadratic error:

$$\underline{\mathbf{g}} = \frac{1}{p}\sum_{\alpha=1}^{p}\left(\underline{\mathbf{x}}^{(\alpha)}\,\underline{\mathbf{w}}^{\top}\underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}^{(\alpha)}y_T^{(\alpha)}\right)$$

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The data in matrix form

Let $\underline{\mathbf{X}}$ represent the observations for all training points:

$$\underline{\mathbf{X}} := \begin{pmatrix} | & | & & | & & | \\ \underline{\mathbf{x}}^{(1)} & \underline{\mathbf{x}}^{(2)} & \dots & \underline{\mathbf{x}}^{(\alpha)} & \dots & \underline{\mathbf{x}}^{(p)} \\ | & | & & | & & | \end{pmatrix} \in \mathbb{R}^{2\times p}$$

and let $\underline{\mathbf{y}}_{\text{True}}$ represent the true labels for all training points:

$$\underline{\mathbf{y}}_{\text{True}} := (y_T^{(1)}, \dots, y_T^{(p)}) \in \mathbb{R}^{1\times p}$$

The qudaratic cost (MSE)

$$\underline{\mathbf{X}} \in \mathbb{R}^{2\times p}$$

$$\underline{\mathbf{y}}_{\text{True}} \in \mathbb{R}^{1\times p}$$

$$\underline{\mathbf{H}} := \frac{1}{p}\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top$$

$$\begin{aligned} E^T_{\text{quadr.}} &= \frac{1}{2p}\left\|\underline{\mathbf{w}}^\top\underline{\mathbf{X}} - \underline{\mathbf{y}}_{\text{True}}\right\|_2^2 \\ &= \frac{1}{2p}\underbrace{\left(\underline{\mathbf{w}}^\top\underline{\mathbf{X}} - \underline{\mathbf{y}}_{\text{True}}\right)}_{1\times p}\underbrace{\left(\underline{\mathbf{w}}^\top\underline{\mathbf{X}} - \underline{\mathbf{y}}_{\text{True}}\right)^\top}_{p\times 1} \end{aligned}$$

with $\underline{\mathbf{y}}_{\text{True}} = \underline{\mathbf{w}}^{*\top}\underline{\mathbf{X}}$:

$$\begin{aligned} &= \frac{1}{2p}\left(\underline{\mathbf{w}}^\top\underline{\mathbf{X}} - \mathbf{w}^{*\top}\underline{\mathbf{X}}\right)\left(\underline{\mathbf{w}}^\top\underline{\mathbf{X}} - \mathbf{w}^{*\top}\underline{\mathbf{X}}\right)^\top \\ &= \frac{1}{2p}\left(\underline{\mathbf{w}} - \mathbf{w}^*\right)^\top\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top\left(\underline{\mathbf{w}} - \mathbf{w}^*\right) \\ &= \frac{1}{2}\left(\underline{\mathbf{w}} - \mathbf{w}^*\right)^\top\underline{\mathbf{H}}\left(\underline{\mathbf{w}} - \mathbf{w}^*\right) \end{aligned}$$

The gradient

$$\underline{\mathbf{X}} \in \mathbb{R}^{2\times p}$$

$$\underline{\mathbf{y}}_{\mathsf{True}} \in \mathbb{R}^{1\times p}$$

$$\underline{\mathbf{H}} := \frac{1}{p}\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top$$

$$\begin{aligned}
\underline{\mathbf{g}} &= \frac{1}{p}\sum_{\alpha=1}^{p}\left(\underline{\mathbf{x}}^{(\alpha)}\,\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}^{(\alpha)}y_T^{(\alpha)}\right)\\
&= \frac{1}{p}\sum_{\alpha=1}^{p}\left(\underline{\mathbf{x}}^{(\alpha)}\,\underline{\mathbf{x}}^{(\alpha)\top}\underline{\mathbf{w}} - \underline{\mathbf{x}}^{(\alpha)}y_T^{(\alpha)}\right)\\
&= \frac{1}{p}\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top\underline{\mathbf{w}} - \underline{\mathbf{X}}\,\underline{\mathbf{y}}_{\mathsf{True}}^\top\right) = \underline{\mathbf{H}}\,\underline{\mathbf{w}} - \frac{1}{p}\underline{\mathbf{X}}\underline{\mathbf{y}}_{\mathsf{True}}^\top
\end{aligned}$$

The gradient

$$\underline{\mathbf{g}} = \frac{1}{p}\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top\underline{\mathbf{w}} - \underline{\mathbf{X}}\,\underline{\mathbf{y}}_{\text{True}}^\top\right)$$

stationary point:

$$\underline{\mathbf{g}} \stackrel{!}{=} \underline{\mathbf{0}} \iff \underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top\underline{\mathbf{w}}^* = \underline{\mathbf{X}}\,\underline{\mathbf{y}}_{\text{True}}^\top$$

$$\underline{\mathbf{w}}^* = \underbrace{\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top\right)}_{\substack{\text{assumes}\\ \text{invertibility}}}^{-1}\underline{\mathbf{X}}\,\underline{\mathbf{y}}_{\text{True}}^\top$$

$\underline{\mathbf{w}}^*$ is a global minimum because...?

The gradient

stationary point:

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$\underline{\mathbf{w}}^*$ is a global minimum because

1. it is a unique solution of $\underline{\mathbf{g}} = \underline{\mathbf{0}}$ and
2. $$\underline{\mathbf{H}} = \frac{1}{p}\sum_{\alpha=1}^{p} \underline{\mathbf{x}}^{(\alpha)}\,\underline{\mathbf{x}}^{(\alpha)\top} = \frac{1}{p}\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top = \underline{\mathbf{C}}$$

   where $\underline{\mathbf{C}}$ is the *empirical covariance matrix* of centered data which is positive definite for invertible $\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top$.

The gradient

stationary point:

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   where $\underline{\mathbf{C}}$ is the *empirical covariance matrix* of centered data which is positive definite for invertible $\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top$.

positive definiteness:

$$\forall \underline{\mathbf{z}} \in \mathbb{R}^2: \quad \underline{\mathbf{z}}^\top \underline{\mathbf{H}}\,\underline{\mathbf{z}} = \frac{1}{p}\,\underline{\mathbf{z}}^\top \underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top \underline{\mathbf{z}} = \frac{1}{p}\left\|\underline{\mathbf{X}}^\top \underline{\mathbf{z}}\right\|_2^2 > 0 \quad \text{iff. } \underline{\mathbf{z}} \neq \underline{\mathbf{0}}$$

L_2 regularization cost:

$$E^R_{[\underline{\mathbf{w}}]} := \frac{1}{2} \sum_{(i,j,v',v)} \left(\mathrm{w}^{v'v}_{ij}\right)^2 = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 = \frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}}$$

$$\frac{\partial}{\partial \underline{\mathbf{w}}} E^R_{[\underline{\mathbf{w}}]} = \ldots?$$

L_2 regularization cost:

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$$\frac{\partial}{\partial \underline{\mathbf{w}}} E^R_{[\underline{\mathbf{w}}]} = \frac{1}{\cancel{2}} \cdot \cancel{2} \cdot \underline{\mathbf{w}} = \underline{\mathbf{w}}$$

The regularized cost function:

$$R_{[\underline{\mathbf{w}}]} = \underbrace{E^T_{[\underline{\mathbf{w}}]}}_{\substack{\text{training} \\ \text{error} \\ \text{e.g. quadratic cost}}} + \underbrace{\lambda E^R_{[\underline{\mathbf{w}}]}}_{\substack{\text{regularization} \\ \text{term}}} \stackrel{!}{=} \min_{\underline{\mathbf{w}}}$$

where

- $E^R \mathrel{\widehat{=}}$ prior knowledge of solution
- λ : regularization parameter

Gradient:

$$R_{[\underline{\mathbf{w}}]} = \underbrace{E^T_{[\underline{\mathbf{w}}]}}_{\substack{\text{training}\\ \text{error}\\ \text{e.g. quadratic cost}}} + \underbrace{\lambda E^R_{[\underline{\mathbf{w}}]}}_{\substack{\text{regularization}\\ \text{term}}} \stackrel{!}{=} \min_{\underline{\mathbf{w}}}$$

$$\underline{\mathbf{g}}^R := ...?$$

Gradient:

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$$\underline{\mathbf{g}}^R := \frac{\partial}{\partial \underline{\mathbf{w}}} R_{[\underline{\mathbf{w}}]} = \frac{\partial}{\partial \underline{\mathbf{w}}} E^T_{[\underline{\mathbf{w}}]} + \frac{\partial}{\partial \underline{\mathbf{w}}} E^R_{[\underline{\mathbf{w}}]}$$

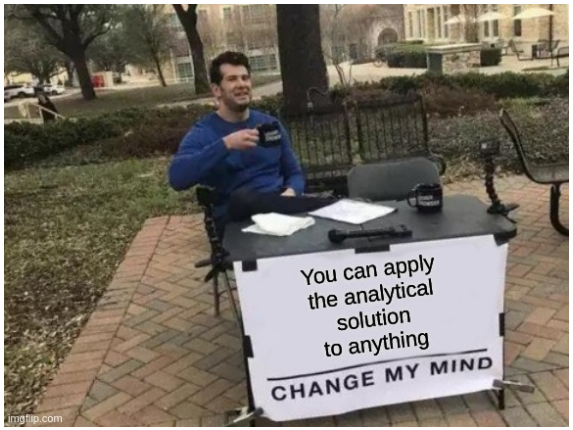
$$= \frac{1}{p}\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top \underline{\mathbf{w}} - \underline{\mathbf{X}}\,\underline{\mathbf{y}}^\top_{\text{True}}\right) + \underline{\mathbf{I}}_N\,\lambda\,\underline{\mathbf{w}}$$

$$\iff \quad \underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top \underline{\mathbf{w}} - \underline{\mathbf{X}}\,\underline{\mathbf{y}}^\top_{\text{True}} + p\lambda\,\underline{\mathbf{I}}_N\,\underline{\mathbf{w}} \stackrel{!}{=} \underline{\mathbf{0}}$$

$$\underline{\mathbf{w}}^* = \underbrace{\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top + p\lambda\,\underline{\mathbf{I}}_N\right)}_{\substack{\text{positive definite and}\\ \text{therefore invertible}}}{}^{-1} \underline{\mathbf{X}}\,\underline{\mathbf{y}}^\top_{\text{True}}$$

Gradient methods

Gradient methods



Q: *In what case(s) do we avoid an analytical solution?*

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- We cannot fit all the data into memory.
- $$\underline{\mathbf{w}}^* = (\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top)^{-1}\underline{\mathbf{X}}\,\underline{\mathbf{y}}_{True}^\top$$

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- We cannot fit all the data into memory.
- $$\underline{\mathbf{w}}^* = (\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top)^{-1}\underline{\mathbf{X}}\,\underline{\mathbf{y}}_{True}^\top$$

  but $\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top$ is not invertible
- no closed-form solution for our model (e.g. MLP with non-linearities)
- Computing the Hessian is too costly. Newton's method also not applicable.
- Non-stationary data. We cannot easily adapt to changes over time.

Alternative: iterative gradient-based optimization

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t - \underbrace{\eta_t \left.\frac{\partial E^T}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}_t}}_{\text{update } \Delta\underline{\mathbf{w}}}$$

with $\eta_t \in \mathbb{R}^+$, then $E^T_{[\underline{\mathbf{w}}_{t+1}]} \leq E^T_{[\underline{\mathbf{w}}_t]}$.

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The behavior of gradient descent can be controlled by modulating

- The "scope" of the gradient. What do we use for deciding on the direction of the update?
- The learning rate schedule. How big of a step do we take in the direction of the gradient?

"Scope" of the gradient

batch	$\dfrac{\partial E^T}{\partial \underline{\mathbf{w}}} = \dfrac{1}{p}\sum_{\alpha=1}^{p} \dfrac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}$	

"Scope" of the gradient

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mini-batch	$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} \approx \frac{1}{\lvert\mathcal{M}\rvert}\sum_{\beta\in\mathcal{M}}^{\lvert\mathcal{M}\rvert}\frac{\partial e^{(\beta)}}{\partial \underline{\mathbf{w}}}$ $\lvert\mathcal{M}\rvert \ll p$ where $\mathcal{M}$ is a set of indices sampled randomly from $\{1,\ldots,p\}$	noisy estimate of E^T, becomes less noisy with larger $\lvert\mathcal{M}\rvert$

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online	$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} \approx \frac{\partial e^{(\beta)}}{\partial \underline{\mathbf{w}}}$ a single random data point $(\underline{\mathbf{x}}^{(\beta)}, \underline{\mathbf{y}}_T^{(\beta)})$	very noisy estimate of E^T

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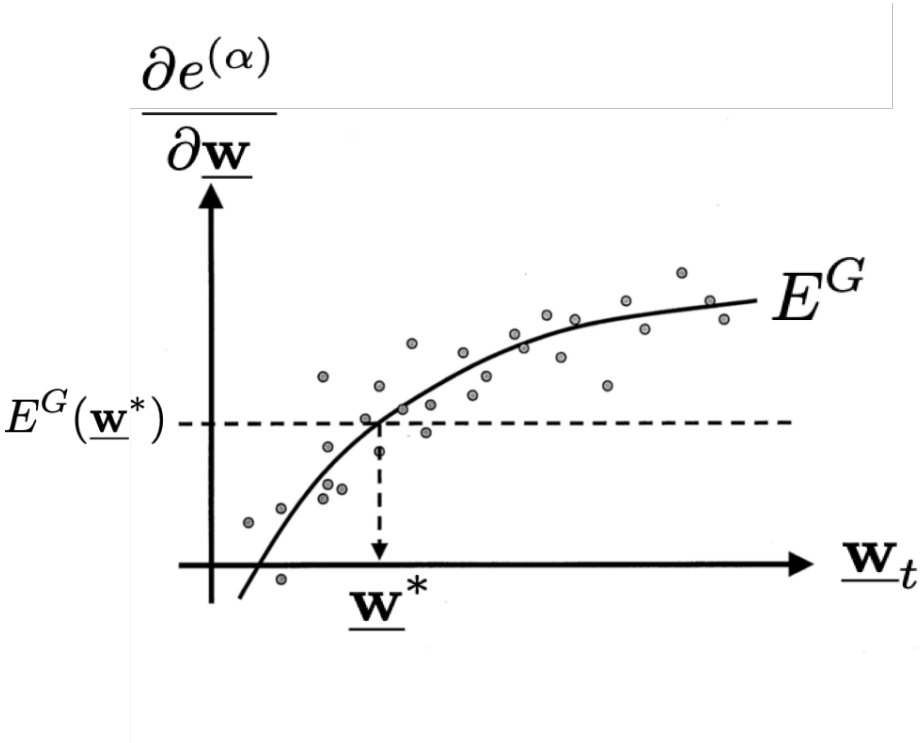
Q: *Which of these can handle non-stationary data?*

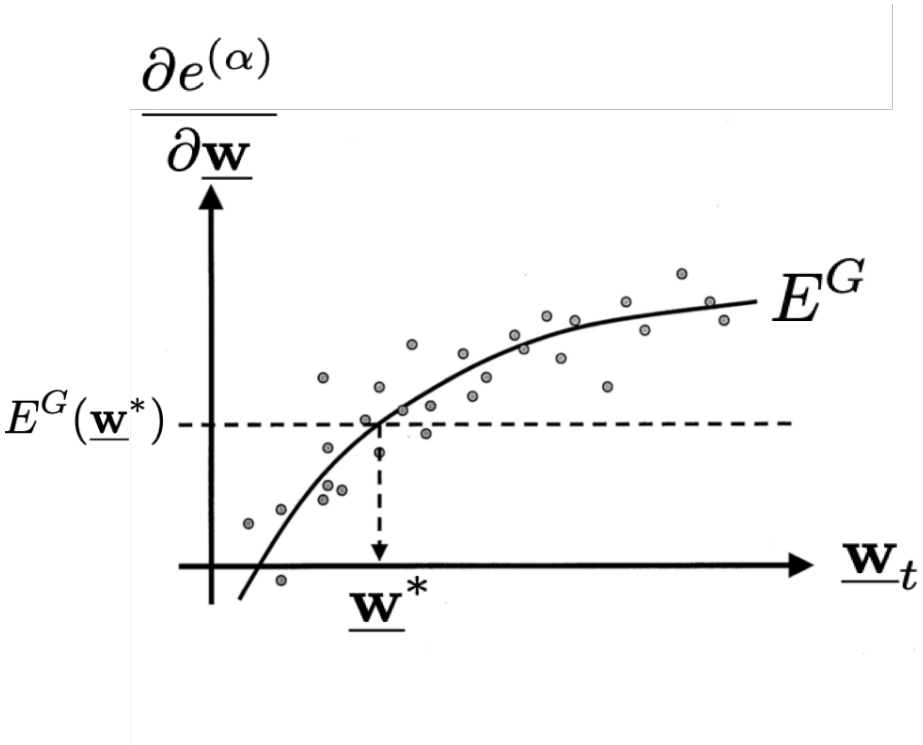
Robbins & Monro condition

(a) Stochastic Approximation
Robbins and Monro (1951)

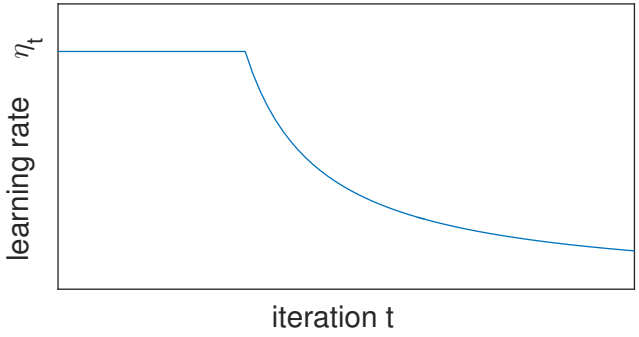


Robbins & Monro condition

(a) Stochastic Approximation
Robbins and Monro (1951)

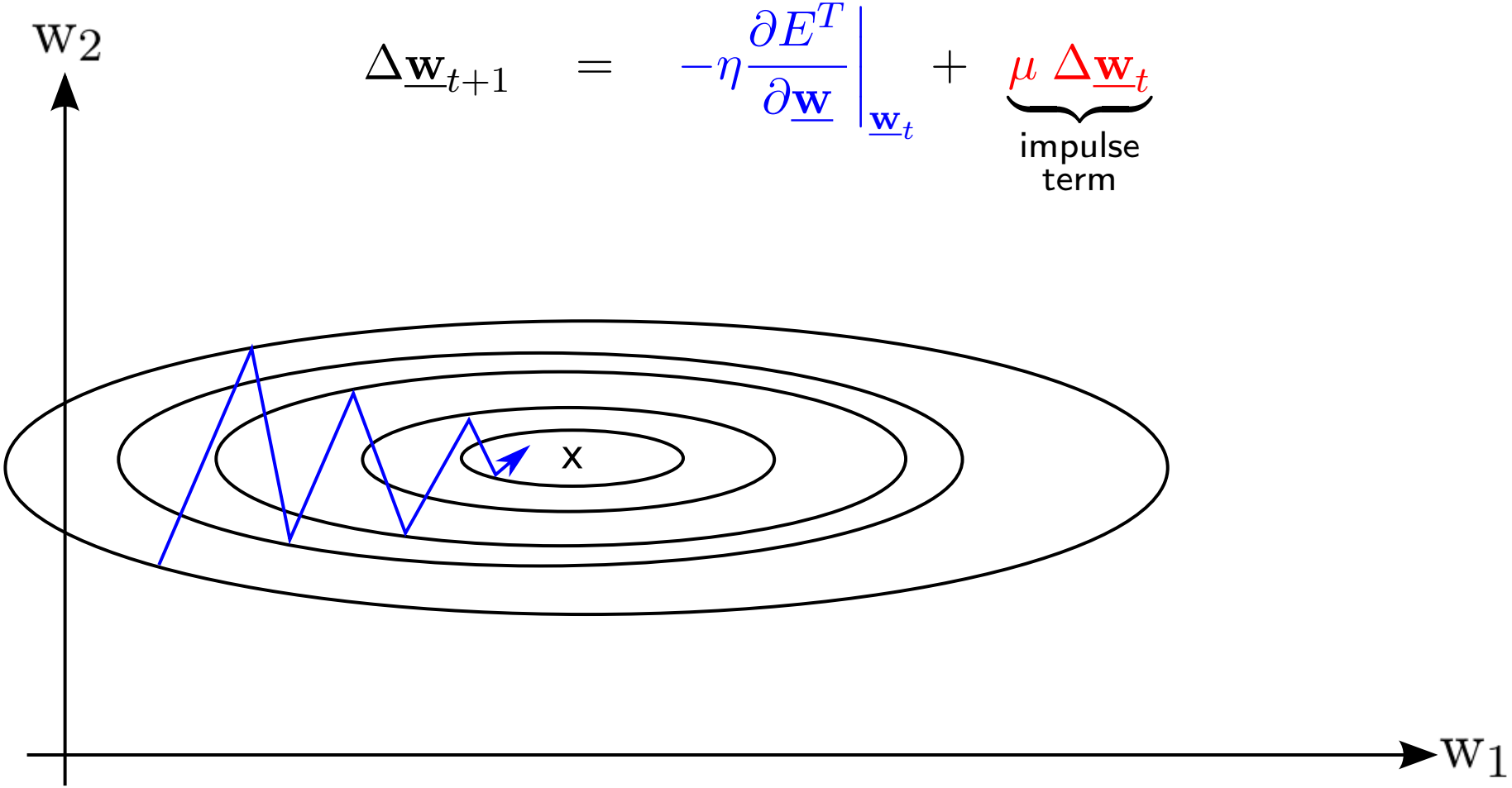


(b) Decaying learning rate



Impulse terms

$$\Delta \underline{\mathbf{w}}_{t+1} = -\eta \frac{\partial E^T}{\partial \underline{\mathbf{w}}}\bigg|_{\underline{\mathbf{w}}_t} + \underbrace{\mu\, \Delta \underline{\mathbf{w}}_t}_{\text{impulse term}}$$



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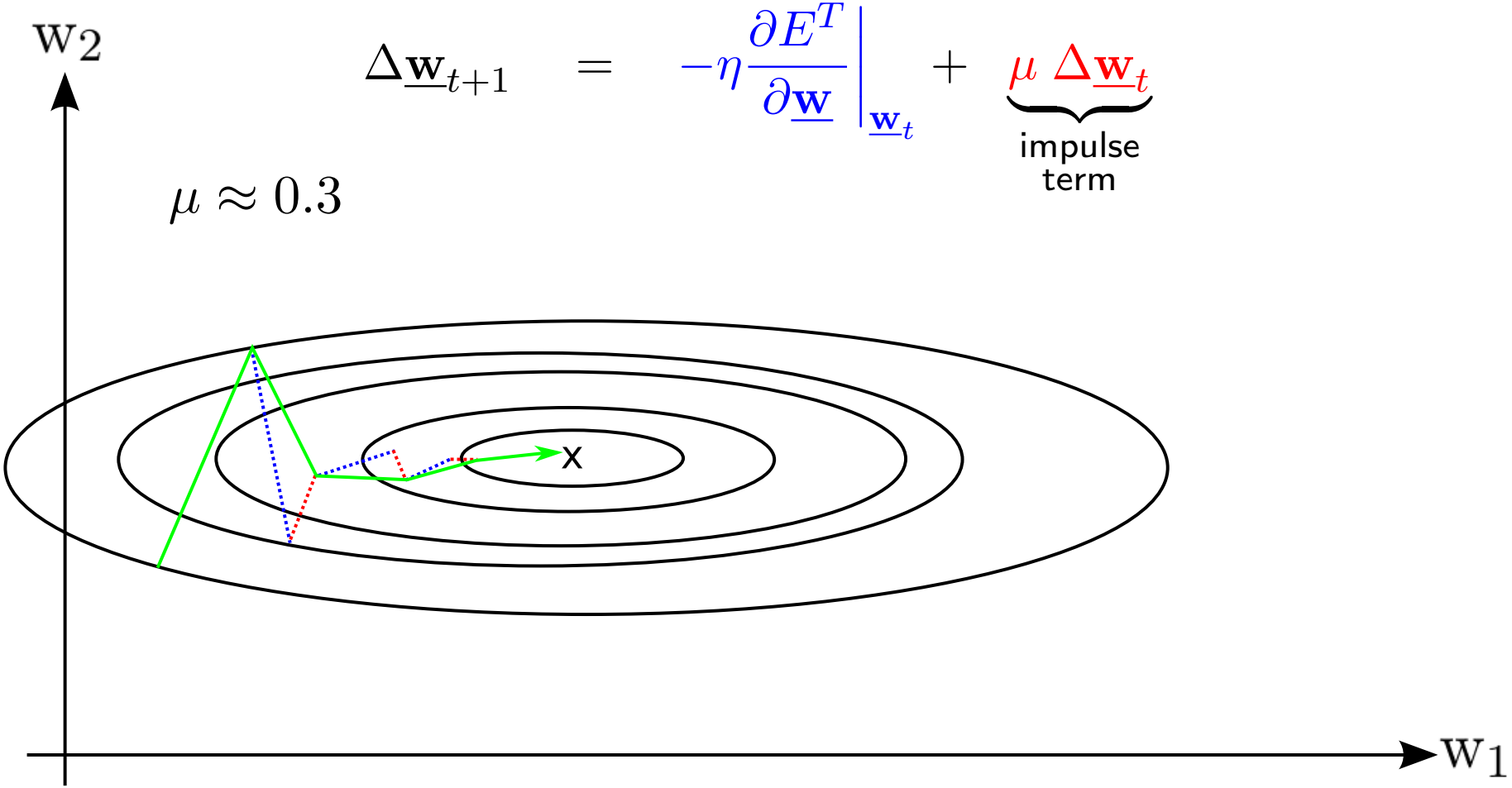
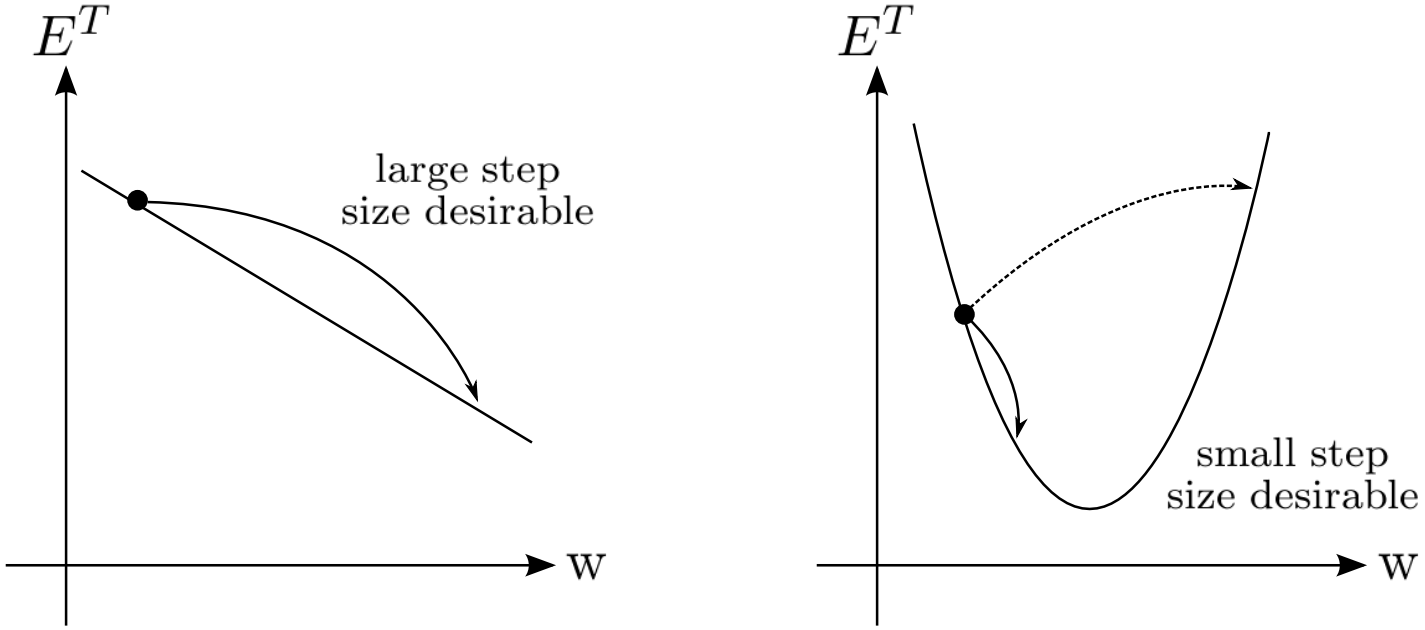
$\mu \approx 0.3$



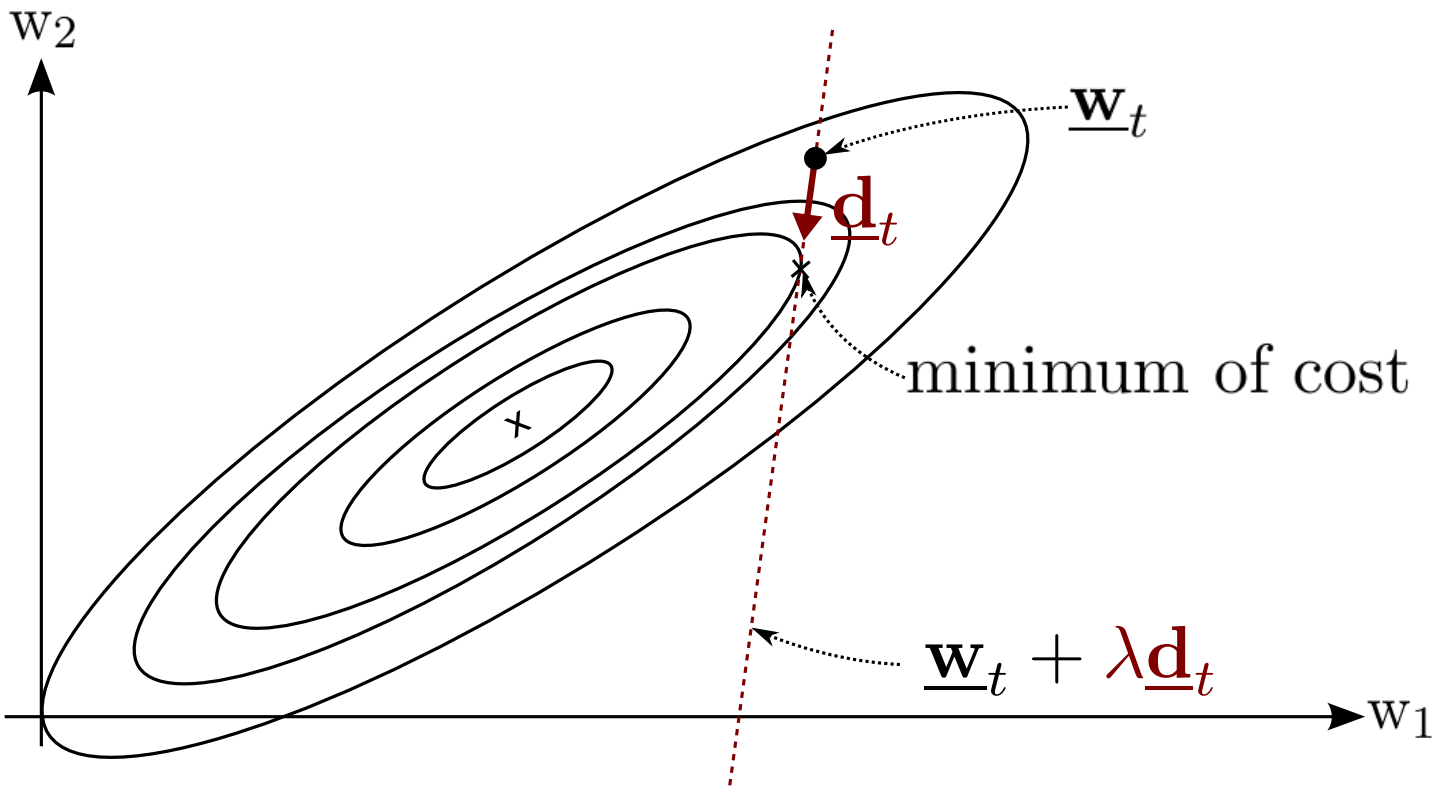
Adaptive step size



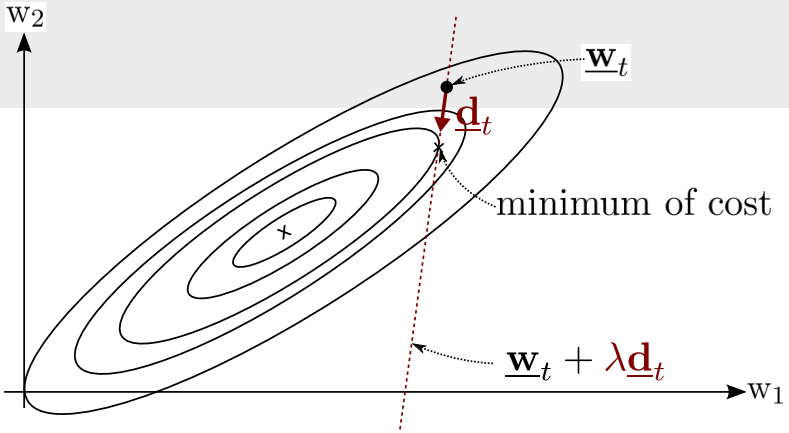
$$\eta_{t+1} = \begin{cases} \rho\eta_t, & \text{if } \Delta E^T < 0, \quad \text{increase step size, if } E^T \downarrow \\ \delta\eta_t, & \text{if } \Delta E^T > 0, \quad \text{decrease step size, if } E^T \uparrow \end{cases}$$

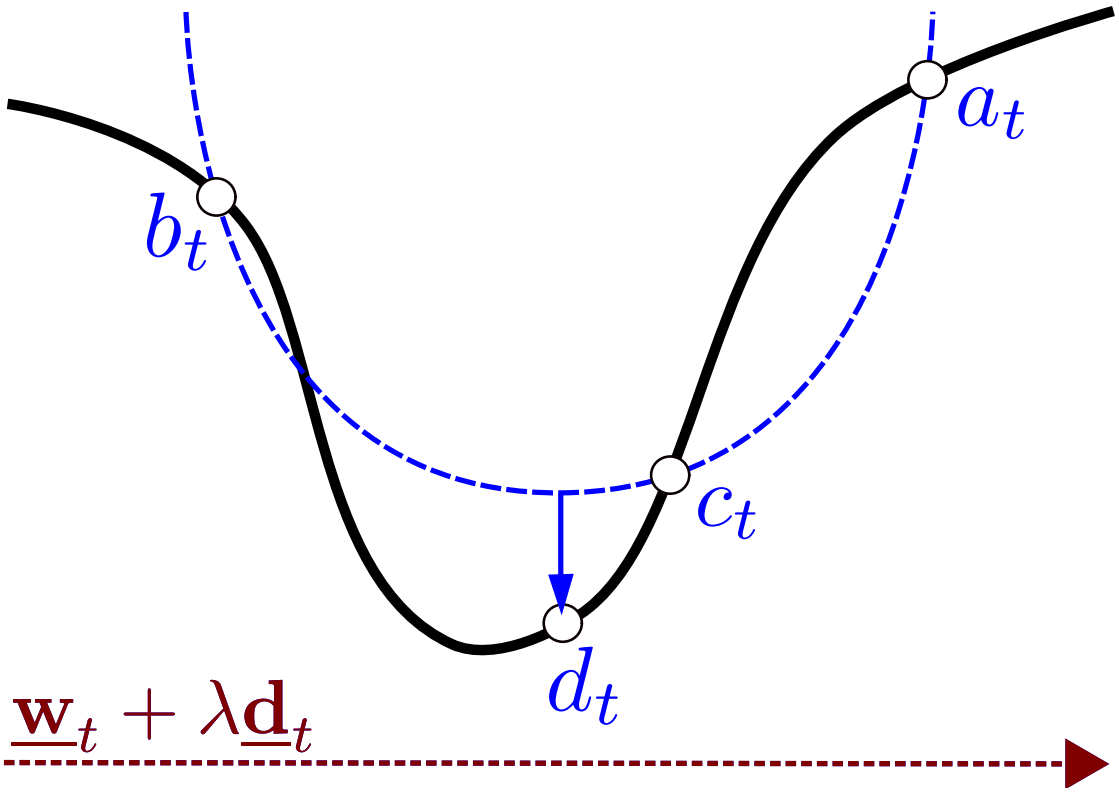
typical values: $\rho = 1.1, \delta = 0.5$

Line search

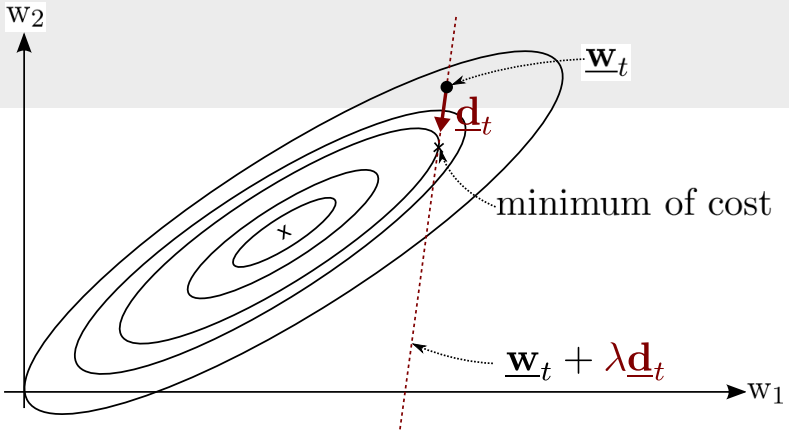

w_2
$\underline{\mathbf{w}}_t$
$\underline{\mathbf{d}}_t$
minimum of cost
$\underline{\mathbf{w}}_t + \lambda \underline{\mathbf{d}}_t$
w_1

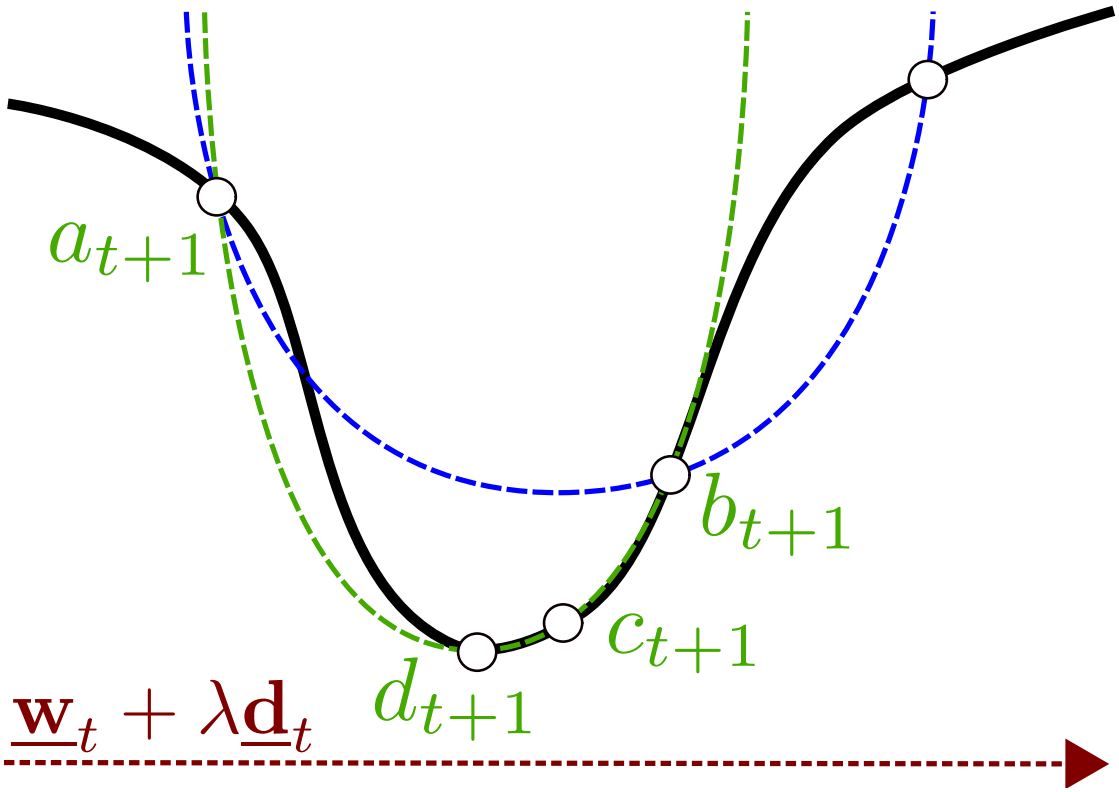
Parabolic interpolation





Parabolic interpolation





Line search

Line search: successive parabolic interpolation

Initialization: a_0, b_0, c_0 (on $\underline{\mathbf{w}}_t + \lambda \underline{\mathbf{d}}_t$); $\quad E^T_{(a_0)}, \quad E^T_{(b_0)} > E^T_{(c_0)}$

while *stopping criterion not fulfilled* **do**

Fit a parabola through the three points a_t, b_t, c_t

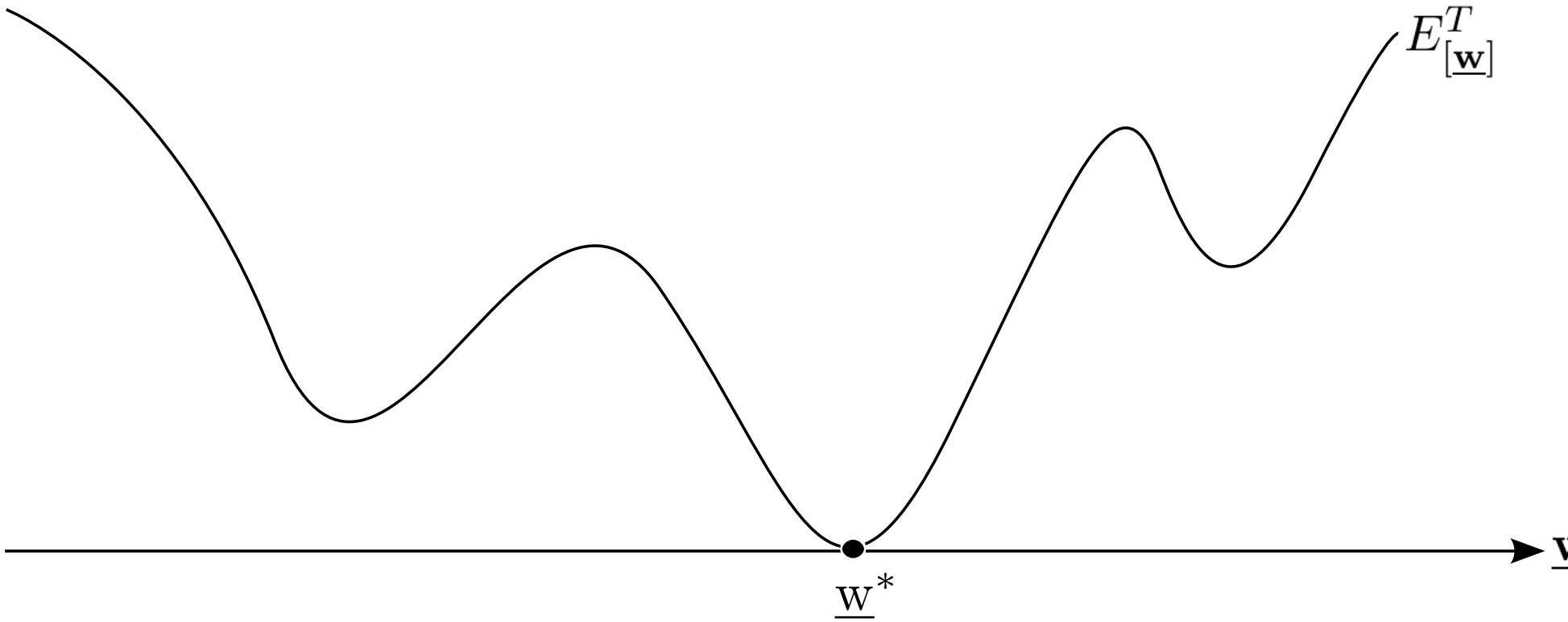
Calculate location d_t of its minimum

Set $c_{t+1} = d_t, \quad b_{t+1} = c_t, \quad a_{t+1} = \begin{cases} a_t, & E^T_{(a_t)} < E^T_{(b_t)} \\ b_t, & \text{else} \end{cases}$

end

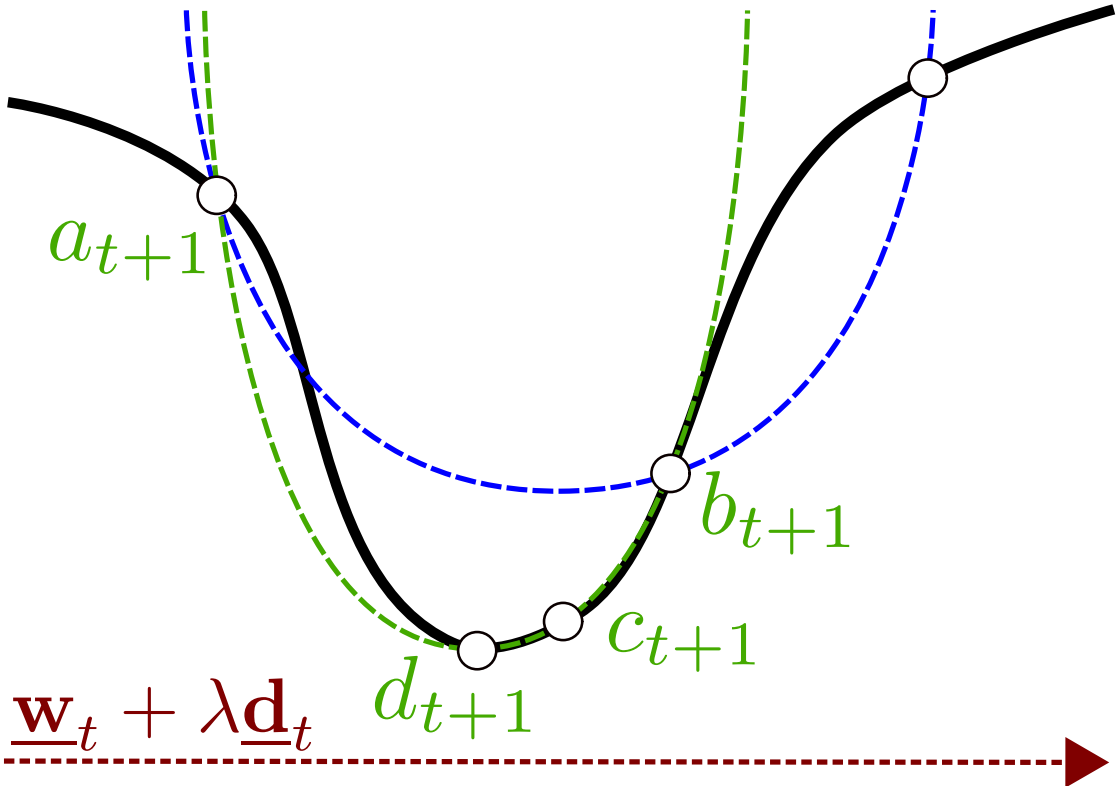
For details and implementation see e.g. Numerical Recipes, 2nd edition, Chapter 10.2.

Line search



Q: *What are potential pitfalls of parabolic interpolation?*

Parabolic interpolation



Compute optimal step size:

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t - \eta_t \underline{\mathbf{g}}_t, \qquad \text{with optimal step size} \qquad \eta_t = \frac{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{H}}\, \underline{\mathbf{g}}_t}\,.$$

Limitations of line search

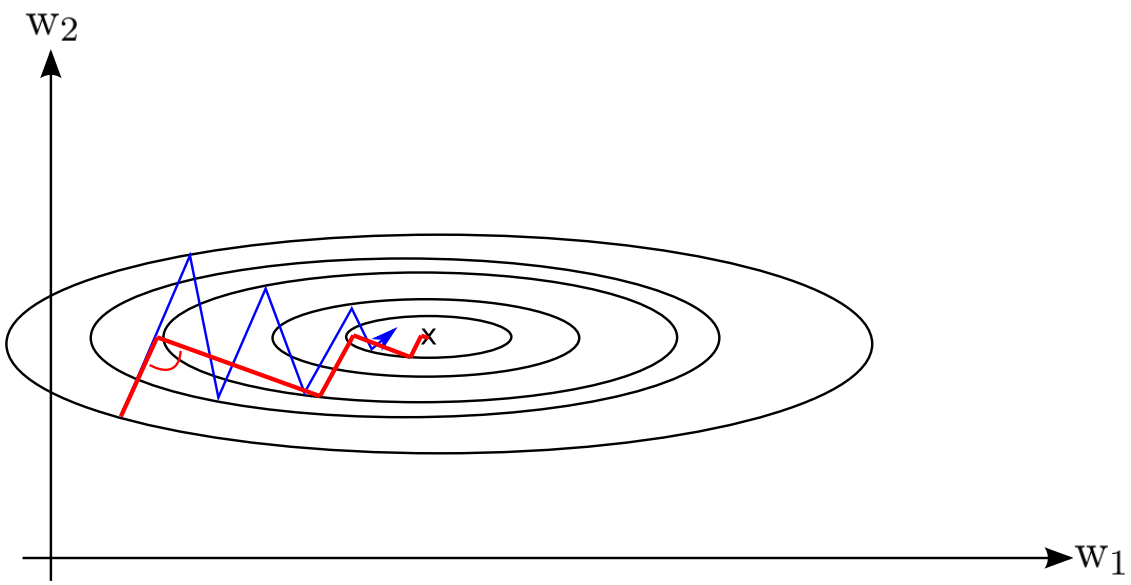


Figure: Weight updates by following line search (red).

Limitations of line search

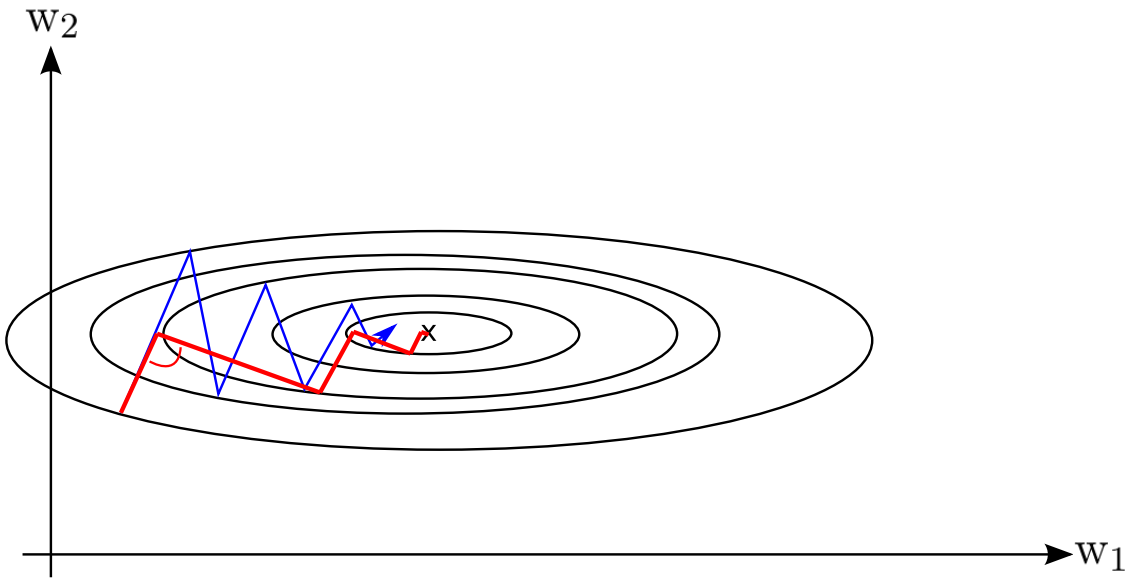
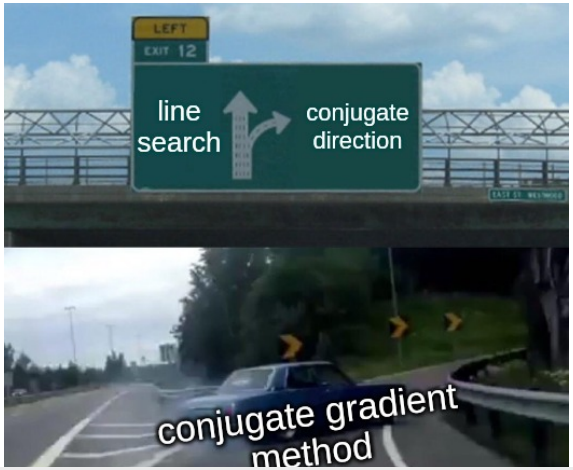
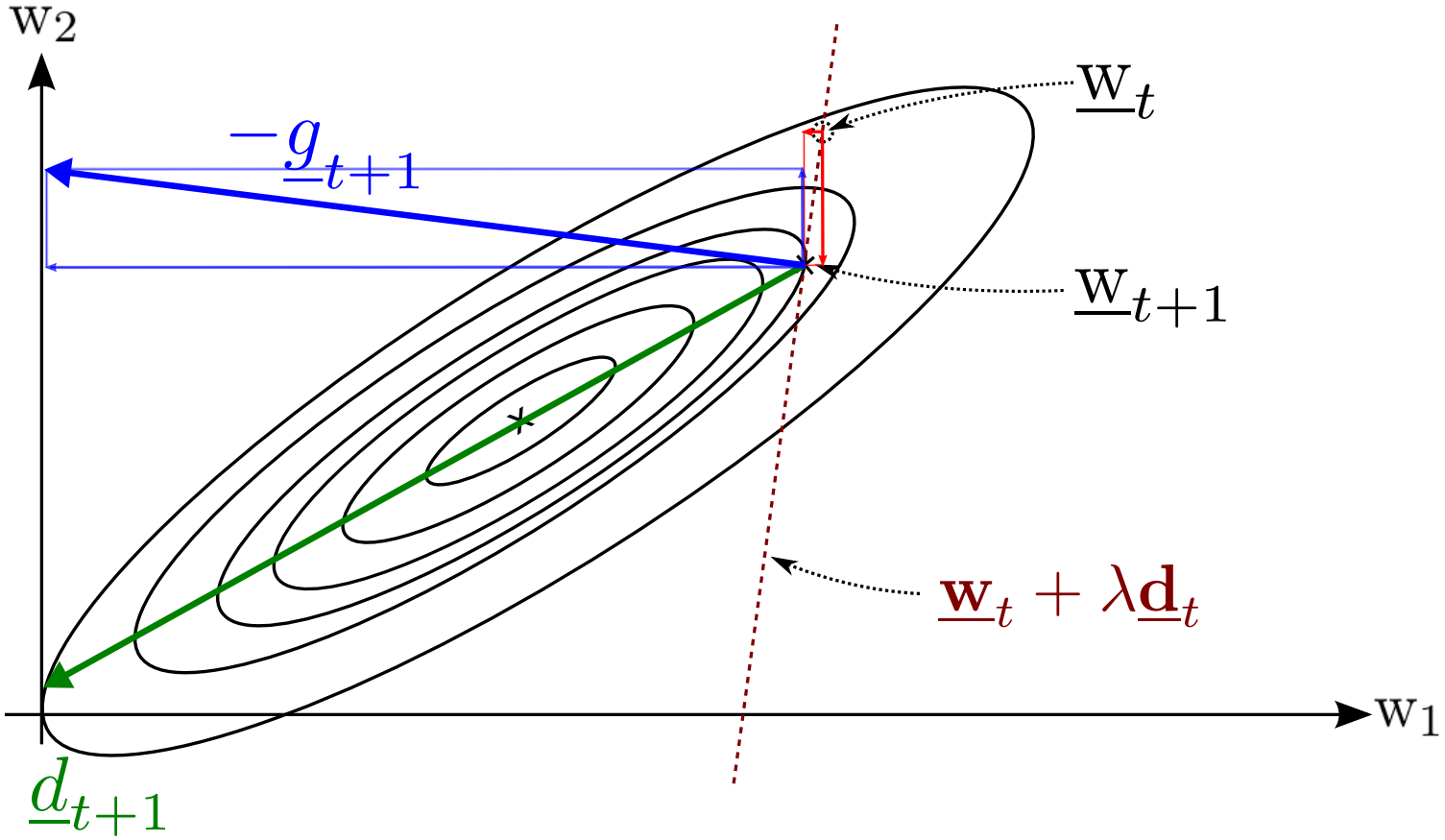


Figure: Weight updates by following line search (red).



Limitations of line search



Conjugate gradient method

$$\underline{\mathbf{w}}_{t+1} \quad = \quad \underline{\mathbf{w}}_t + \eta_t \, \underline{\mathbf{d}}_t$$

Don't undo progress of previous search directions $\underline{\mathbf{d}}_t$, $\underline{\mathbf{d}}_{t-1}$.
In other words:
The component of the gradient parallel to the old direction $\underline{\mathbf{d}}_t$ should remain zero along the new direction $\underline{\mathbf{d}}_{t+1}$

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Initialization: $\underline{\mathbf{w}}, \quad \underline{\mathbf{d}} = -\underline{\mathbf{g}}$

while *stopping criterion not fulfilled* **do**

Minimize E^T along $\underline{\mathbf{d}}$ using line-search $\rightarrow$ new $\underline{\mathbf{w}}$

Calculate the new conjugate direction $\rightarrow$ new $\underline{\mathbf{d}}$

end

Adaptive momentum: the Polak-Ribiere rule

$$\underline{\mathbf{d}}_{t+1} = -\overbrace{\left.\frac{\partial E^T}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}_{t+1}}}^{\underline{\mathbf{g}}_{t+1}} + \beta_t \, \underline{\mathbf{d}}_t$$

$$\underbrace{\beta_t = \frac{\underline{\mathbf{g}}_{t+1}^\top \left(\underline{\mathbf{g}}_{t+1} - \underline{\mathbf{g}}_t\right)}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}}_{\text{Polak-Ribiere rule}}$$

OR

$$\underbrace{\beta_t = -\frac{\underline{\mathbf{g}}_{t+1}^\top \underline{\mathbf{g}}_{t+1}}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}}_{\text{Fletcher-Reeves}} .$$

Conjugate gradient descent algorithm

Initialization: $\underline{\mathbf{w}}, \quad \underline{\mathbf{d}} = -\underline{\mathbf{g}}$

while *stopping criterion not fulfilled* **do**

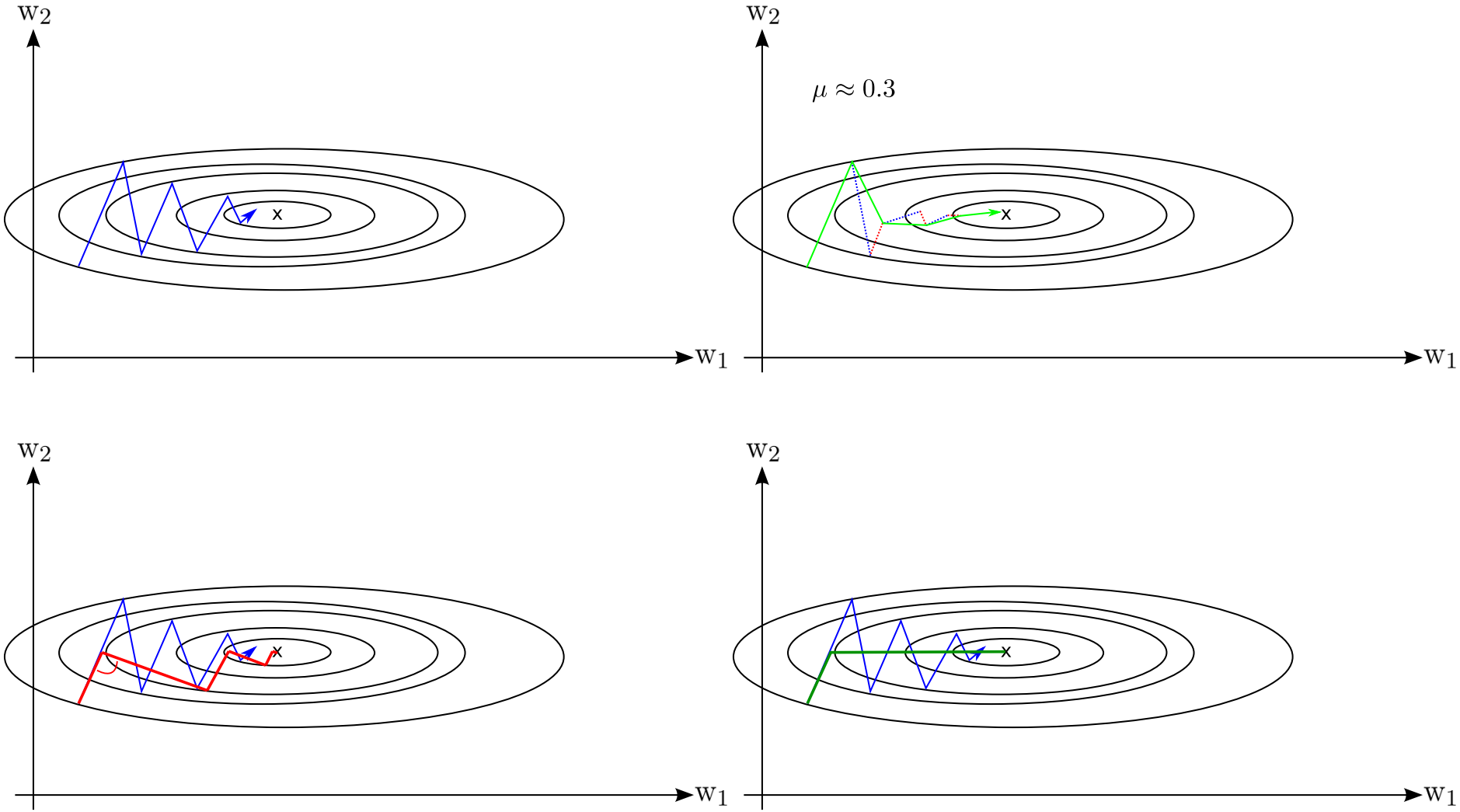
Minimize E^T along $\underline{\mathbf{d}}$ using line-search $\rightarrow$ new $\underline{\mathbf{w}}$

Calculate the new conjugate direction $\rightarrow$ new $\underline{\mathbf{d}}$

end

- Subsequent search directions lose conjugacy
  - search direction should be reset to the steepest descent direction at least every N iterations (in practice apply heuristic, e.g. reset every 5 steps)

Comparison



References I

Herbert Robbins and Sutton Monro. A Stochastic Approximation Method. *The Annals of Mathematical Statistics*, 22(3):400 – 407, 1951. doi: 10.1214/aoms/1177729586. URL https://doi.org/10.1214/aoms/1177729586.